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Oct. - Dec. 2024

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Honeyypot & ASCII Art Playground

Key Objectives



Threat analysis

Be exposed to real life attack.



Practise defending measurements

Get experience with hand-on defending measurement.



Explore intersection of Hacking and art

Would there be any similarities between hacking and art that I could dig into?



Explore hacker psychology

After listening blogs about Marcus Hutchins: what are hackers thinking when they hack? How may they react to ASCII cats?

spa cat

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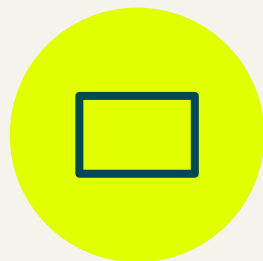
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Timeline



Launched an EC2

- Security rule: ssh only allows MY IP address
- Monitored by Amazon CloudWatch (Network, CPU utilization...)

Set up server

- Confirmed that firewall, ssh connection are working
- Allowed ping
- Set up ASCII arts in files in the server
- monitored login attempts with `/var/log/auth.log` and monitored command lines with `./bash_history`

Brute force attacked

- port 22 permission denied
- Nmap: ssh closed
- 3.5G + inbound network
- 6.5M + outbound network

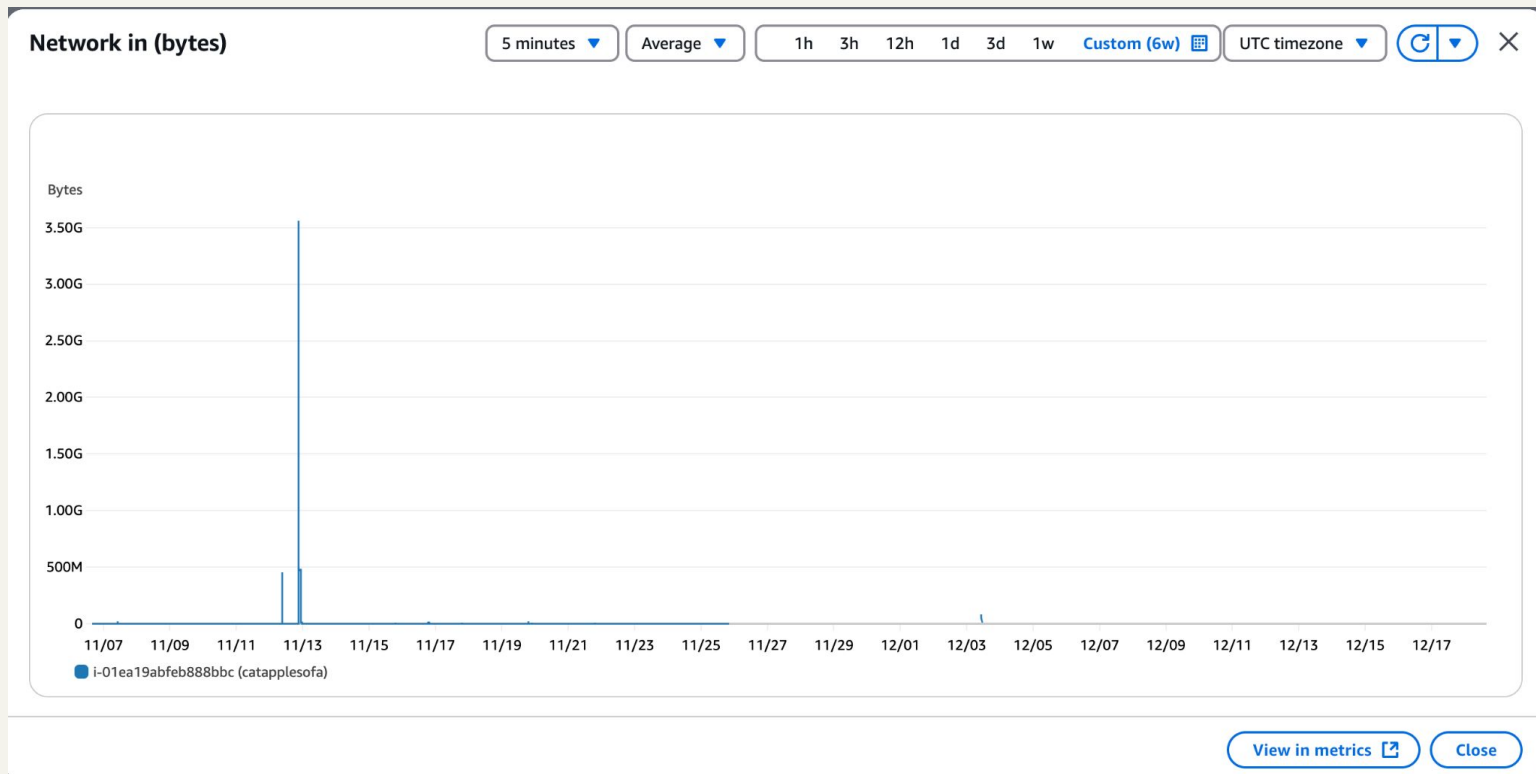
Reinforcement: 2222 and vpn

- Turned down ping
- Moved ssh from port 22 to port 2222
- Put ec2 behind a vpn to restrict inbound network connections

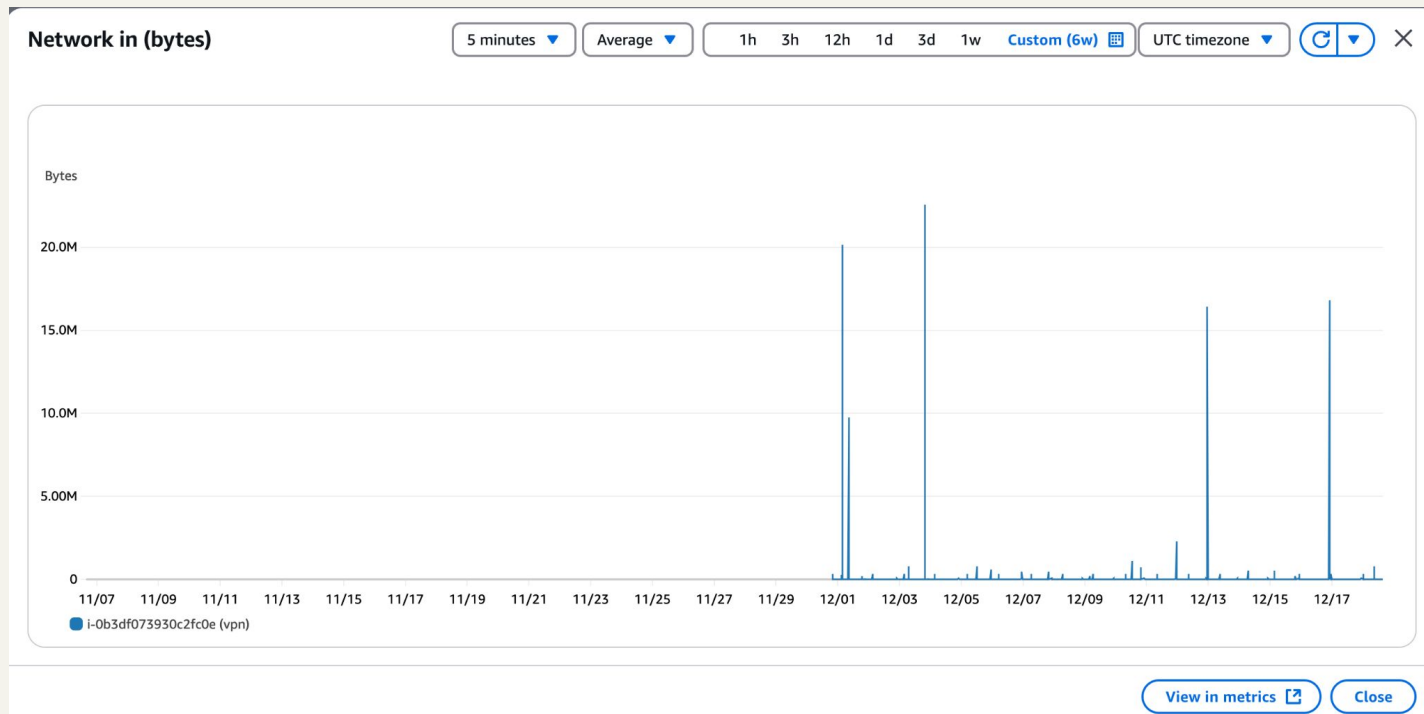
cappuccino cat

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First server x_x



VPN + Port 2222



What I learned

From this experience:

Real world attacks VS in class attacks

Real world attacks are very likely to be conducted by zombie machines with programs, instead of manually. These attacks are in large scale and fast.

Post attack

Important to have monitoring and control methods outside of the server (to stop a server after it is attacked).

Online study resources

- chatgpt: explaining low level concepts
- Amazon tutorial: high level tools (eg. session manager)
- Forums: re:)Post, how2shot.com : troubleshooting

noodle cat

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Thank You!

Special thanks to
Brandan Keaveny (DataEthics)
Ming Chow (Tufts University)
Mina Zhou (Amazon)

Chatgpt
AWS
Stackoverflow
...

end of semester cat

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